

## Assignment Submission

Dear Hiring Team,

Thank you for giving me the opportunity to work on this assignment. I enjoyed building the project and learned a lot during the development and deployment process.

### Technical Approach & Architectural Decisions

I built the project using a full-stack architecture.

- **Frontend:** React.js with Tailwind CSS for a clean and responsive user interface.
- **Backend:** Node.js and Express.js to build REST APIs.
- **Database:** MySQL hosted on Railway.
- **Deployment:** Frontend and Backend deployed on Render, Database hosted on Railway.
- **API Communication:** Axios was used for communication between the frontend and backend.

The application follows a modular structure where the frontend, backend, and database are separated, making the project easier to maintain and scale.

---

### Key Features

Some features I am most proud of are:

- Customer can create support tickets.
- Dashboard displaying all support tickets.
- Search and filter tickets by status.
- Update ticket status (Open, In Progress, Closed).
- Add investigation notes to tickets.
- Fully deployed application with a cloud-hosted MySQL database.
- Clean and responsive user interface.

---

### Challenges Faced & Solutions

During deployment, I faced several challenges.

- The backend initially could not connect to the Railway MySQL database because it was using Railway's private network. I solved this by enabling Public Network access and updating the database connection variables in Render.

- The frontend was still calling the local backend (localhost) after deployment. I fixed this by using Vite environment variables (VITE\_API\_URL) and redeploying the frontend.

These issues helped me better understand deployment, environment variables, and cloud database connectivity.

---

### **Improvements with More Time**

If I had additional time, I would implement:

- User authentication and login.
  - Role-based access (Admin and Support Agent).
  - Email notifications for ticket updates.
  - File attachment support.
  - Ticket priority and category management.
  - Dashboard charts and analytics.
  - Pagination for large numbers of tickets.
  - Unit testing and API documentation.
- 

### **Deployed Application**

#### **Frontend (Live Demo):**

<https://support-crm-frontend-n3y7.onrender.com/>

#### **Backend API:**

<https://support-crm-backend-ep5k.onrender.com/>

---

### **GitHub Repository**

[https://github.com/sujalsargar/SUPPORT\\_CRM](https://github.com/sujalsargar/SUPPORT_CRM)

### **Google Drive Demo Link**

[https://drive.google.com/file/d/1dFUg\\_6Vb\\_O7dGHYd2YdyNadRze7EY1Yv/view?usp=drivesdk](https://drive.google.com/file/d/1dFUg_6Vb_O7dGHYd2YdyNadRze7EY1Yv/view?usp=drivesdk)

---

### **Youtube Demo Video**

<https://youtu.be/cYbCDQoCZFk?si=lnkLV-jZ5R38Y3ax>

---

Thank you for reviewing my submission. I appreciate your time and consideration. This assignment gave me valuable hands-on experience in building, deploying, and troubleshooting a full-stack application. I look forward to your feedback.

Kind regards,

**Sujal Sargar**

**sargarsujal4341@gmail.com**

**+91 9082877083**